PHYSICS 2
0580

GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2024 **ORDINARY LEVEL**

Subject Title	**Physics**
Subject Code No.	**0580**
Paper No.	**2**

Duration: Two and a Half Hours

Answer ALL questions.

Section I is designed to be answered in 1 hour and Section II in 1½ hours.

You are advised to divide your time accordingly.

In section II answer EITHER a, b and c OR d, e, and f of each question

For your guidance the approximate mark for each part of a question is indicated in brackets.

You are reminded of the necessity for good English and orderly presentation of your answers.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

Where necessary, assume:

- *the acceleration due to gravity on the surface of the earth, $g = 10\ m\ s^{-2}$*
- *the speed of light in air, $c = 3.0 \times 10^{8}\ m\ s^{-1}$*
- *the charge on an electron, $e = 1.6 \times 10^{-19}\ C$*

Calculators are allowed.

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SECTION I (1 hour)
Answer ALL questions

1. (a) Define pressure and state its unit of measurement **(2 marks)**
Explain why a sharpened edge knife cuts meat easily, when a constant force is applied. **(2 marks)**
(b) Calculate the pressure water exerts at a depth of $6\,m$ in a pond given that the density of water is $1000\,kg\,m^{-3}$. **(2 marks)**

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2. (a) Distinguish between contact and non-contact forces. **(2 marks)**
(b) Explain why an object weighs more when taken from the equator of the earth to the poles **(2 marks)**

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3. The table in figure 1 contains part of the electromagnetic spectrum arranged in decreasing order of wavelength.

Visible light	X	Y	Gamma rays

Figure 1

(a) Identify the radiations X and Y and state how each can be detected. **(4 marks)**
(b) Give two differences between electromagnetic waves and sound waves. **(2 marks)**
(c) Differentiate between transverse and longitudinal waves. **(2 marks)**

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4. Figure 2 shows a uniform plank of negligible mass and of length $2\,m$, on a support, A, at its midpoint. Two weights of $60\,N$ and $80\,N$ are attached to it.

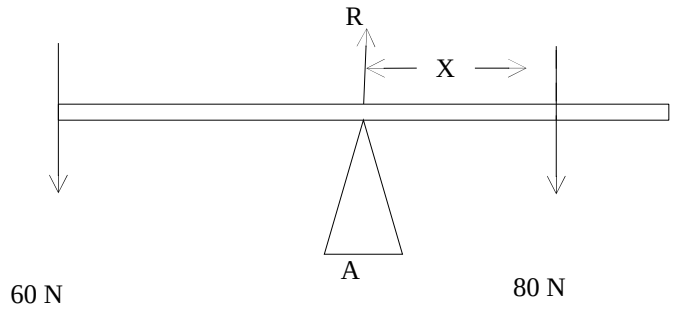


Figure 2

(a) State two conditions necessary for the system to be in equilibrium. **(2 marks)**
Calculate:
(b) the distance, x, of the $80\,N$ object from the support when the system is in equilibrium. **(2 marks)**
(c) the force, R, on the plank by the support **(2 marks)**

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5. (a) (i) Define interference. **(1 mark)**
(ii) Name the two types of interference. **(2 marks)**
(iii) Explain what happens to the resultant amplitude of each type named in (ii) above. **(2 marks)**
(b) Figure 3 shows how a stationary wave is formed on a string tied at both ends A and B.

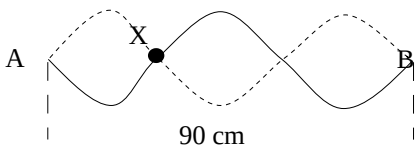


Figure 3

(i) Calculate the wavelength of the stationary wave. **(2 marks)**
(ii) State and explain what happens to the small object placed at X**.** **(2 marks)**

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6. Figure 4 shows a simple pulley system used by a builder to lift a bag of cement of mass 50 kg to a height of $4\,m$ by applying an effort of $300\,N$.

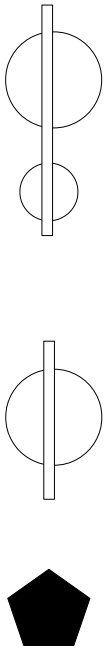


Figure 4

(a) What is the velocity ratio of the machine? **(1 mark)**
(b) Calculate the weight of the bag of cement. **(2 marks)**
(c) Calculate the mechanical advantage of the pulley system. **(2 marks)**
(d) Calculate the efficiency of the pulley system. **(2 marks)**

SECTION II (1 ½ hours)

Answer ALL questions choosing, EITHER a, b and c OR d, e, and f of each question.

Answer EITHER 7 a, b and c

7. (a) (i) Define acceleration and state its units **(2 marks)**
   (ii) What is the acceleration of a body moving with a constant velocity? **(1 mark)**
   (iii) Describe an experiment to determine the acceleration of a moving body using the ticker tape timer. Your description should include:
   - a labelled diagram of the experimental set-up.
   - the procedure you will follow.
   - the data you will collect
   - how you will use the data to obtain the acceleration of the body
   - any precaution you will take to ensure better results. **(7 marks)**

   (b) A car travelling with a constant velocity of $20\,ms^{-1}$ for 10 s is brought to rest in 2 s after applying the brakes.
   (i) What is meant by constant velocity? **(1 mark)**
   (ii) Draw a velocity-time graph for the above motion. **(3 marks)**
   (iii) Calculate the distance covered by the car during the whole journey. **(2 marks)**

   (c) A small lead ball falling from a height attains terminal velocity as it falls.
   (i) Define terminal velocity. **(1 mark)**
   (ii) Draw the small lead ball and show all the forces acting on it as it falls. **(3 marks)**

OR 7 d, e, and f

7. (d) (i) Define refractive index. **(2 marks)**
   (ii) State what happens to the speed of light when a ray of light moves from air to water. **(1 mark)**
   (iii) Describe an experiment to determine the refractive index of water using the real and apparent depth method. Your description should include:
   - a labelled diagram of the experimental set-up.
   - the procedure you will follow.
   - the data you will collect
   - how you will use the data to obtain the refractive index of water
   - any precaution you will take to ensure better results. **(7 marks)**

   (e) A prism can be used to produce an impure spectrum of light on a screen.
   (i) What is meant by an impure spectrum of light? **(1 mark)**
   (ii) Draw a ray diagram to show how the impure spectrum is formed on a screen. **(3 marks)**
   (iii) Explain how a pure spectrum can be formed. **(2 marks)**

   (f) A convex lens is used to produce a virtual enlarged image of an object in a school lab.
   (i) What is a virtual image? **(1 mark)**
   (ii) Draw a ray diagram to show how the lens forms the image of the object. **(3 marks)**

Answer EITHER 8 a, b and c

8. (a) When an un-calibrated liquid-in-glass thermometer is placed in pure melting ice, the length of the mercury thread is $5\,cm$. When it is placed in steam from boiling water under standard atmospheric pressure, the length of the thread is $45\,cm$.

(i) Define thermometric property. **(2 marks)**

(ii) State the thermometric property of the liquid-in-glass thermometer. **(1 mark)**

(iii) What is the name given to the mercury level on the thermometer when it is placed in steam? **(1 mark)**

(iv) Calculate the temperature for which the length of the thread will be $25\,cm$. **(2 marks)**

(b) In an experiment to determine the boiling point of a liquid, the temperature of the liquid, θ, and the corresponding times, t, were recorded as the liquid was heated for a period of $20\,minutes$. The results obtained are shown on the table below.

$\theta/°C$	27	65	83	95	95	95
t/min	0	4	8	12	16	20

(i) Define boiling point. **(1 mark)**

(ii) Plot a graph of temperature, θ, on the y-axis against time, t, on the x-axis. **(5 marks)**

(iii) Determine the boiling point of the liquid from your graph. **(2 marks)**

(iv) Give one reason why the boiling point can be different from the standard value. **(1 mark)**

(c) An electric heater rated $900\,W$ is used to heat $2\,kg$ of water at $30\,°C$ in 5 minutes. Calculate:

(i) the quantity of heat energy supplied to the water during this time. **(3 marks)**

(ii) the change in temperature of the water given that the specific heat capacity of water is $4200\,J\,kg^{-1}°C^{-1}$. **(2 marks)**

OR 8 d, e and f

8. (d) In an experiment to determine the count rate of a radioactive substance, a student records a count of $20\,counts\ per\ second$ and $18\,counts\ per\ second$ respectively before and after the experiment when there was no actual source present in front of the GM tube. When a radioactive carbon was placed in front of the GM tube, a count rate of $239\,counts\ per\ second$ was recorded.

(i) Define background radiation. **(1 mark)**

(ii) State one source of background radiation. **(1 mark)**

(iii) Calculate the background count rate for this experiment. **(1 mark)**

(iv) Calculate the actual count rate of the radioactive carbon. **(2 marks)**

(e) In an experiment to determine the half-life of a radioactive substance, the count rates of the substance and the corresponding times were recorded as the substance decays for a period of $200\,hours$. The results obtained are shown on the table below.

$Count\ rate/C\,s^{-1}$	1800	1200	900	640	450	340
$Time/hrs$	0	20	40	60	80	100

(i) Define half-life. **(1 mark)**

(ii) Plot a graph of count rate on the y-axis against time on the x-axis. **(5 marks)**

(iii) Use the graph to determine the half-life of the substance. **(2 marks)**

(iv) Give one reason why some of the plotted points do not fall on the line. **(1 mark)**

(f) An unstable radioactive substance ${}^{56}_{22}X$ decays to a stable substance Y by emitting an alpha and a beta particle.

(i) Write the equation for the decay process with the correct symbols of the alpha and beta particles. **(2 marks)**

(ii) Determine the mass number and the proton number of the substance Y **(4 marks)**

Answer EITHER 9 a, b and c

9. (a) You are provided with a positively charged rod X, and two charged spheres A and B. One of the spheres is positively charged while the other is negatively charged.
   (i) What is a charged material? **(1mark)**
   (ii) Describe test that you would carry out using the charged rod to identify the charges on the two spheres. **(4 marks)**
   (iii) Draw the electric field pattern around the positively charged sphere. **(2 marks)**

   (b) Figure 5 shows a circuit diagram operating three 6Ω bulbs connected as shown.

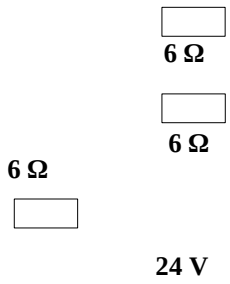


Figure 5

Calculate:
(i) the combined resistance of the resistors. **(2 marks)**
(ii) the current flowing through the circuit. **(2 marks)**
(iii) the total power dissipated by the bulbs. **(2 marks)**
(iv) state two advantages of connecting bulbs in parallel. **(2 marks)**

(c) A wire has a resistance of 10Ω. A similar wire of the same material is twice as long but of the same thickness.
(i) Compare the resistances of the two wires. **(3 marks)**
(ii) State and explain one other factor which affects the resistance of the wire. **(2 marks)**

OR 9 d, e and f

9. (d) You are provided with a bar magnet, M, and two other bars, X and Y. One a bar magnet and the other a magnetic material.
   (i) What is a magnet? **(1 mark)**
   (ii) Describe test that you would carry out using the magnet M to identify the other bars. **(4 marks)**
   (iii) Draw the magnetic field pattern around the bar magnet. **(2 marks)**

   (e) Figure 6 shows a simple diagram of a transformer used in a welding workshop.

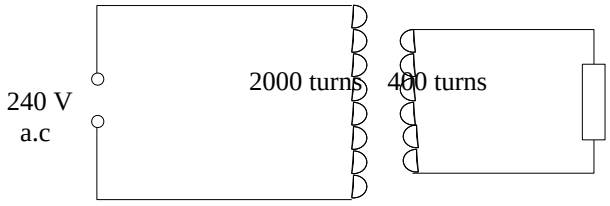


Figure 6

Calculate:
(i) the output voltage V_s. **(2 marks)**
(ii) the power output if the current in the secondary circuit is $8\ A$. **(2 marks)**
(iii) the efficiency of the transformer given that the input current is $2\ A$. **(2 marks)**
(iv) state two sources of energy loses in transformers. **(2 marks)**

(f) A diode is connected to the secondary coil of the transformer in figure 6.
(i) Draw two cycles of the input voltage and two cycles of the output voltage. **(3 marks)**
(ii) State and explain the function of the diode. **(2 marks)**

STOP
GO BACK AND CHECK YOUR WORK